From RNN and LSTM to Transformers and AI Agents

How Modern AI Systems Generate Knowledge

Course	203.3763 — Orchestration of AI Agents
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Assembled by a CrewAI multi-agent pipeline and typeset with LuaLaTeX.

Abstract

This article traces the evolution of sequence modelling in modern artificial intelligence: from recurrent neural networks (RNNs) and their long short-term memory (LSTM) refinement, through the attention mechanism and the Transformer architecture, to the multi-agent systems that orchestrate them. We explain why recurrent models were originally needed, how gating addressed the vanishing-gradient problem, and why scaled dot-product attention enabled parallel, long-range context modelling. We then connect attention and context to agentic systems, using the CrewAI framework as an example of role-based orchestration. The document itself was assembled by such a pipeline, and serves as a worked example of how a team of cooperating agents can research, write, illustrate, and typeset an academic PDF end to end.

Contents

Chapter 1

Introduction

Sequence data — language, audio, time series — is everywhere, and modelling it well has driven much of modern AI. This article follows that thread from early recurrent models to the Transformer and on to multi-agent orchestration. The core mechanism that unifies the later chapters, scaled dot-product attention, is given in Equation 4.1, and the conceptual trade-offs between the models are summarised in Table 6.1 and Figure 4.1.

Chapter 2

Recurrent Neural Networks (RNNs)

Recurrent networks process a sequence one step at a time, maintaining a hidden state that carries information forward [6]. This makes them a natural fit for variable-length input, but the same step-by-step recurrence makes long-range dependencies hard to learn: gradients propagated through many time steps tend to vanish or explode [2], limiting how far back the model can effectively remember.

Chapter 3

Long Short-Term Memory (LSTM)

The LSTM addresses the vanishing-gradient problem with a gated cell state [5]. Input, forget, and output gates regulate what is written to, retained in, and read from the cell, letting error signals flow across many steps without decaying. Gated recurrent units later offered a lighter variant of the same idea [3]. These models substantially extended the usable memory horizon, yet remained fundamentally sequential.

Chapter 4

Transformers and Attention

Attention let a model align directly with any position in the input [1], and the Transformer removed recurrence entirely in favour of self-attention [7]. The scaled dot-product attention operation is:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{4.1}$$

Each query in Q is scored against every key in K; dividing by $\sqrt{d_k}$ keeps the scores from saturating the `softmax`, whose weights then form a context-aware combination of the values V. Because every position attends to every other in parallel, the maximum dependency path is constant rather than linear in the sequence length, as Figure 4.1 illustrates. This design also underpins large pre-trained encoders such as BERT [4].

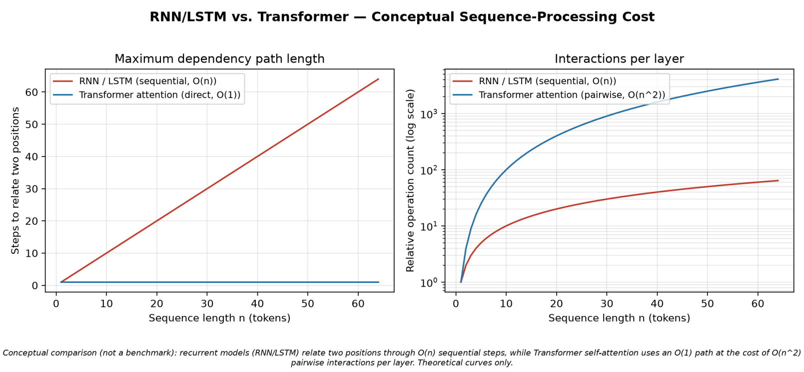

Figure 4.1: Conceptual comparison of the maximum dependency path length for recurrent models (linear in sequence length) versus Transformer self-attention (constant). Theoretical illustration, not a benchmark.

Chapter 5

AI Agents and CrewAI

Attention gives a model context within a single forward pass; agentic systems extend the same idea across time and tools. The CrewAI framework organises large-language-model agents into roles with explicit goals and shared context, running them as a sequential or hierarchical process. The output of one agent becomes the context of the next, much as this document was produced. See the CrewAI documentation for the framework's API.

Chapter 6

Comparison Table

Table 6.1 contrasts the mechanisms discussed above. It restates established, course-aligned definitions; it contains no measured results.

Table 6.1: Conceptual comparison of recurrent models, the Transformer, and CrewAI agent orchestration.

System	**Core mechanism**	**Long-range**	**Parallelism**
RNN	Recurrent hidden state	Weak (vanishing grad.)	Low
LSTM	Gated cell state	Improved via gating	Low
Transformer	Self-attention	Strong (direct access)	High
CrewAI agents	Roles + shared context	Carried by task context	Orchestrated

Chapter 7

Bilingual Chapter — Hebrew and English (BiDi)

This chapter demonstrates bidirectional (BiDi) typesetting: left-to-right English mixed with right-to-left Hebrew, handled by `polyglossia` under LuaLaTeX. The Hebrew term for “attention” is קשב, and “neural network” is רשת עצבית. A full Hebrew paragraph follows: מנגנון הקשב (noitnetta) מאפשר למודל להתמקד בחלקים הרלוונטיים של הקלט. ארכיטקטורת הטרנספורמר מבססת את כל החישוב על מנגנון זה, ובכך מתגברת על מגבלות הזיכרון של רשתות נשנות כגון NNR ו-MTSL. מערכות סוכנים כמו IAwerC מרחיבות את רעיון ההקשר אל מעבר למודל בודד. The two scripts coexist in one chapter with correct directionality.

Chapter 8

Conclusion

From RNNs to LSTMs to Transformers, each step lengthened the reach of context; multi-agent orchestration carries that context across tools and tasks. The references below ground every architectural claim in its original source.

References

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